

Implementing Zhang's Six-Frame Algorithm Using CUDA

Althea Zyrle Arceta, Ryan Onil Barrion, Antonio Gabriel Mendoza, Jose Tristan Tan and Roger Luis Uy,
Adviser, De La Salle University

Abstract

In RNA sequencing, the alignment phase is critical for interpreting RNA-seq data and often takes significant time. Hence, sequence alignment algorithms are often utilized in order to speed up the alignment. With the growth of biological sequence data, efficient computational methods for sequence alignment have become increasingly important. Zhang's six-frame alignment approach is critical for aligning RNA sequencing reads. However, its sequential implementation hinders efficiency, especially on big datasets. This study aims to parallelize Zhang's six-frame alignment algorithm utilizing NVIDIA's CUDA framework, which uses the SIMT (Single Instruction, Multiple Threads) execution model to improve computing speed. The study entails creating a sequential implementation, adapting it for the SIMT paradigm, and deploying it with CUDA. The parallelized algorithm's performance will be compared to its sequential version and an existing SIMD (Single Instruction, Multiple Data) implementation, with tests performed on synthetic and real-world datasets such as *Scylla serrata* and *Drosophila melanogaster*.

Keywords

DNA-Protein Alignment, RNA-seq, CUDA, SIMT, Frameshift Alignment

I. INTRODUCTION

RNA-seq is widely used for studying gene expression in non-model organisms, with applications ranging in agriculture [1], aquaculture [2], ecology [3], and the environment [4]. This technology entails extracting RNA, converting it to complementary DNA (cDNA), and sequencing it to determine gene expression patterns. Aligning sequencing reads to a reference genome, or transcriptome allows for the quantification of gene expression, identification of differentially expressed genes, and detection of novel transcripts and isoforms. This alignment phase is critical for interpreting RNA-seq data and often takes a significant amount of time. Hence, this phase frequently incorporates multiple-frame translation or sequence alignment algorithms [5].

The advancement of next-generation sequencing technologies has accelerated RNA-seq development, opening doors to exciting new applications. These include examining transcriptomes (the whole set of RNA molecules in a cell) and developing novel tools such as long-read RNA-seq and direct RNA-seq. Furthermore, advances in computer tools for data processing and novel approaches to RNA-seq add greatly to our ever-deepening understanding of RNA biology [6].

Sequence alignment techniques like Smith-Waterman [7], BLAST [8], and BWA [9] are tools developed for RNA-seq data analysis. These techniques align sequencing reads with reference sequences, ensuring precise mapping even in insertions, deletions, and mismatches. For example, the Smith-Waterman approach enables local alignment, effectively detecting regions of high similarity within larger sequences, which is critical for aligning RNA-seq reads with different exon-intron architectures. Similarly, Burrows-Wheeler Aligner (BWA) and Bowtie are designed for high-throughput sequencing data, providing quick and efficient alignment, which is crucial given the vast number of RNA-seq data. Using these techniques, researchers may correctly assess gene expression, find alternative splicing events, and discover novel transcripts, thereby improving our understanding of the transcriptome and gene control.

Another form of sequence alignment is the Frameshift alignment. The frameshift alignment is an algorithm that compares DNA read sequences to the reference proteome based on the start codon's position and strand, considering frameshift errors. An example of Frameshift alignment is Zhang's three-frame alignment algorithm [10] which is a global alignment that compares DNA read sequences based on the strand and position of the start codon to the reference proteome. It accounts for frameshift defects but not for introns in the DNA sequence, making the algorithm appropriate for comparing DNA sequences that suffer from sequencing errors with an amino acid or protein sequence database. The three frames are obtained by translating the DNA sequence based on the start codon's locations. For example, in the sequence AGTAG, Frame 1 would have AGT translate to serine, Frame 2 translate GTA to valine, and Frame 3 would have TAG to signify a stop codon. Three frames are crucial for comparing all possible DNA and protein sequence translations. If only one frame is used, frameshifts are impossible because insertions and deletions in the DNA sequence are ignored.

However, Zhang's three-frame alignment only compares one input DNA sequence to the protein sequence, despite the fact that a DNA molecule has both positive and negative strands. Hence, it is critical to test both strands to assess the feasibility of

the input. Zhang's six-frame alignment algorithm [11] utilizes 3 more frames that will be obtained from the reverse complement of the DNA sequence (a total of 6 frames). It computes the optimal alignment of the six frames by using the three-frame alignment algorithm twice, accounting for both the negative and positive strands.

However, a challenge occurs when data production in bioinformatics outpaces computing capabilities. Sequence alignment, a fundamental procedure in bioinformatics analysis, is becoming more expensive as sequence databases expand exponentially [12]. Thus, a need for alternative solutions to sequential implementations arises to keep pace with the growth of data used in the algorithms. To combat this issue, researchers often utilize parallel computing techniques to enhance the execution of these algorithms, leading to faster analyses and the capability to process larger datasets. SIMT (Single Instruction, Multiple Threads) and SIMD (Single Instruction, Multiple Data) are two parallel computing paradigms that enable effective parallel processing [13].

Single instruction multiple data (SIMD) achieves parallelism by utilizing several processing units that perform a single instruction on each element in a data set. It provides a range of effective approaches for rapidly constructing algorithms and is employed in most applications requiring large data processing [14]. A SIMD implementation of Zhang's six-frame algorithm [11] using AVX2 employs a staircase-like method to remove the data dependency inherent in the three-frame alignment algorithm [10], allowing the algorithm to be implemented in the SIMD paradigm, resulting in a threefold increase in execution speed over the sequential version.

On the other hand, Graphics Processing Units (GPUs) use the Single Instruction, Multiple Threads (SIMT) execution model for parallel computing. This method allows a large number of distinct threads to execute the same instruction concurrently while operating on diverse data [16]. This makes GPUs efficient for activities that can be divided into multiple independent operations.

Popular frameworks such as NVIDIA's CUDA (Compute Unified Device Architecture) and OpenCL enable programmers to leverage the capability of SIMT on GPUs. CUDA, specifically designed for NVIDIA GPUs, offers a full programming environment for GPGPU processing. It frequently performs best due to its close connection with NVIDIA hardware [15]. On the other hand, OpenCL takes a more general-purpose approach, allowing code to be portable across multiple SIMT-compatible hardware platforms, including AMD and Intel GPUs. However, when targeting specific hardware, this portability may sacrifice some speed compared to CUDA. Since achieving maximizing performance is a priority for this study, we will utilize CUDA for our SIMT implementation.

As far as the researchers are concerned, there are still no direct implementations regarding a frame alignment algorithm in SIMT, although there is a possibility of ongoing development in this research area. Hence, this research aims to implement Zhang's Six-Frame Alignment Algorithm in the SIMT paradigm. To achieve this, the research outlines the following specific objectives:

- 1) To modify the existing methodology of the algorithm to fit the data-parallel nature of the SIMT model seamlessly.
- 2) To develop a sequential implementation of Zhang's Six-Frame Alignment Algorithm.
- 3) To implement Zhang's Six-Frame Alignment Algorithm with the SIMT execution model using CUDA.
- 4) To evaluate the developed parallel implementation, including comparing its performance to the sequential version and existing SIMD implementation, and then analyzing the results to compare and contrast the performance gains achieved by the proposed parallelization approach.

II. REVIEW OF RELATED LITERATURE

1. Local Alignment Methods

Local alignment methods in bioinformatics align the regions of DNA or amino acid sequences with the highest density matches. Several algorithms and tools allow you to find the local alignments, such as the Smith-Waterman algorithm, Zhang's Six-Frame Alignment algorithm, BLAST, and FASTA. The challenge presented by these algorithms is that they use a dynamic programming approach, which can be computationally expensive on large datasets [18]. Several parallelization techniques on the recurrence Smith-Waterman Algorithm, FASTA, and Six-Frame alignment were addressed by [11][18][25] address this problem by using their machine's GPU and examining their execution times based on the number of GPU cores utilized, or by implementing these algorithms using parallel computing platforms such as OpenMP, Linda, and SIMD. The results of these studies show that the parallelization of these algorithms leads to an acceleration of speedup time and efficient memory use, even with large datasets.

A different way to address these challenges produced by the dynamic programming approach for the local alignment method is by using a configurable integrated circuit such as a Field-programmable Gate Array (FPGA). A parallelization technique for

the BLAST tool was implemented on FPGA[22]. The method used in the study of Vouzis, P. D., & Sahinidis, N. V. (2010) implementing the Gapped BLAST algorithm along with their two-hit method in the FPGA. Their hardware design was able to have high-performance and efficient FPGA implementations. A different approach to speeding up the accelerating the algorithm was conducted by Guong X. et al. (2012) where they developed a sequence alignment method called “GPU-BLAST”, which was a version of the popular NCBI-BLAST, and was implemented on the GPU. Their GPU implementation was able to speed up mostly between 3 to 4 times compared to the NCBI-BLAST algorithm [20].

2. Global Alignment Methods

Global alignment in bioinformatics aligns two DNA or protein sequences over their entire length. Similar to Local alignment methods, some algorithms compete for the most optimal alignment such as Zhang’s Three-Frame alignment algorithm and the NeedleMan-Wunsch algorithm which uses a dynamic programming approach. Implementations of these algorithms were made using parallel computing platforms such as CUDA, OpenMP, and SIMD[11][23][24], and compared to their sequential approach implemented using programming languages such as C and Java. Lim, S.V. et al. (2021) developed a modified version of Zhang’s Three Frame algorithm by reducing the number of computations of the algorithm’s recurrence equation.[10]. However, their traceback algorithm was implemented sequentially. Implementations of these algorithms on parallel computing platforms demonstrated a speedup in computation compared to their sequential counterpart despite having the traceback algorithm run sequentially.

III. SCOPE AND LIMITATIONS

The RNA sequencing process consists of multiple components, and sequence alignment is one such component. This project only concerns itself with sequence alignment, particularly frame alignment. This research also addresses several key points related to parallelizing a frame alignment algorithm. First, the algorithm to be implemented is Zhang’s six-frame algorithm. While other algorithms are available, the study will only focus on the aforementioned algorithm.

The algorithm will also be implemented using CUDA's Single Instruction, Multiple Thread (SIMT) execution model. To do this, the present methodology will be modified to meet the SIMT model's data-parallel structure. While other data parallelization methods exist, this study will focus only on the SIMT execution model using CUDA.

Finally, an evaluation of the proposed parallel implementation will be conducted. Testing will be performed on synthetic and real-world datasets to ensure the accuracy and consistency of the alignments produced by the parallelized algorithm. This evaluation compares its performance against the sequential version as well as the results of the previous SIMD implementation. By analyzing the results, this research quantifies the performance benefits of the proposed parallelization approach.

IV. PROCESS OVERVIEW DIAGRAM

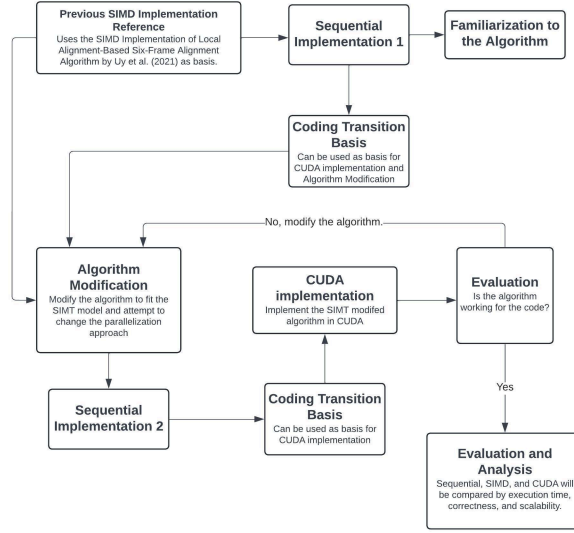


Fig. 1: Proposed Project Process

V. SIGNIFICANCE OF THE STUDY

Despite the advanced approach for optimization of the recurrence of the algorithms through a dynamic programming approach, the sequential execution of each instruction of the program cannot meet the demand for efficiency and scalability as the dataset grows in size and increases in sequence reading [12]. The findings of this study will contribute to the knowledge regarding parallelization approaches using SIMT and CUDA in the field of Computer Science. Furthermore, the findings of this study will provide acceleration speed up on algorithms regarding sequence alignment, which provides a powerful way to compare related DNA or protein sequences in Bioinformatics.

REFERENCES

- [1] S. Wang, W. Li, Y. Liu, H. Li, Y. Ma, and Z. Zhang, "Comparative transcriptome analysis of shortened fruit mutant in woodland strawberry (*Fragaria vesca*) using RNA-Seq," *Journal of integrative agriculture*/Journal of Integrative Agriculture, vol. 16, no. 4, pp. 828–844, Apr. 2017, doi: [https://doi.org/10.1016/s2095-3119\(16\)61448-x](https://doi.org/10.1016/s2095-3119(16)61448-x).
- [2] Cláudia Raposo de Magalhães, Kenneth Sandoval, Ferenc Kagan, Grace McCormack, Denise Schrama and Raquel Carrilho, "Transcriptomic changes behind *Sparus aurata* hepatic response to different aquaculture challenges: An RNA-seq study and multiomics integration," *PloS one*, vol. 19, no. 3, pp. e0300472–e0300472, Mar. 2024, doi: <https://doi.org/10.1371/journal.pone.0300472>.
- [3] M. Kamitani, A. J. Nagano, M. N. Honjo, and Hiroshi Kudoh, "A Survey on Plant Viruses in Natural Brassicaceae Communities Using RNA-Seq," *Microbial ecology*, vol. 78, no. 1, pp. 113–121, Oct. 2018, doi: <https://doi.org/10.1007/s00248-018-1271-4>.
- [4] Koichi Akashi, Hiromi Kumamaru, Yasuyuki Ohkawa, Hirokazu Saiwai, Hisakata Yamada, Kensuke Kubota, and Kazu Kobayakawa, "Direct isolation and RNA-seq reveal environment-dependent properties of engrafted neural stem/progenitor cells," *Nature communications*, vol. 3, no. 1, Oct. 2012, doi: <https://doi.org/10.1038/ncomms2132>.
- [5] X. Guan and E. Uberbacher, "Alignments of DNA and protein sequences containing frameshift errors," *Computer applications in the biosciences : CABIOS*, Feb. 1996, <https://www.semanticscholar.org/paper/Alignments-of-DNA-and-protein-sequences-containing-Guan-Uberbacher/98ec4f29b58f4d86ebcc58499e91d8abf2e451b>.
- [6] R. Stark, M. Grzelak, and J. Hadfield, "RNA sequencing: the teenage years," *Nature reviews. Genetics*, vol. 20, no. 11, pp. 631–656, Jul. 2019, doi: <https://doi.org/10.1038/s41576-019-0150-2>.
- [7] T. F. Smith and M. S. Waterman, "Identification of common molecular subsequences," *Journal of Molecular Biology*/Journal of molecular biology, vol. 147, no. 1, pp. 195–197, Mar. 1981, doi: [https://doi.org/10.1016/0022-2836\(81\)90087-5](https://doi.org/10.1016/0022-2836(81)90087-5).
- [8] S. F. Altschul, W. Gish, W. Miller, E. W. Myers, and D. J. Lipman, "Basic local alignment search tool," *Journal of Molecular Biology*/Journal of molecular biology, vol. 215, no. 3, pp. 403–410, Oct. 1990, doi: [https://doi.org/10.1016/s0022-2836\(05\)80360-2](https://doi.org/10.1016/s0022-2836(05)80360-2).
- [9] H. Li and R. Durbin, Fast and accurate short read alignment with Burrows-Wheeler Transform. *Bioinformatics*, 25:1754-60, 2010, Sourceforge.net, 2010. <https://bio-bwa.sourceforge.net/>.
- [10] Z. Zhang, W. R. Pearson, and W. Miller, "Aligning a DNA Sequence with a Protein Sequence," *ResearchGate*, Feb. 1997. https://www.researchgate.net/publication/13942672_Aligning_a_DNA_Sequence_with_a_Protein_Sequence.
- [11] Stanley Vincent Lim, Steven Edward Lim, Carlos Louis Ting, Aaron Eldrich Wong, and Roger Luis Uy, "SIMD Implementation of Modified Zhang's Three-Frame Alignment Algorithm," 2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), Nov. 2021, doi: <https://doi.org/10.1109/hnicem54116.2021.9731997>.
- [12] S. Sarkar, Gaurav Ramesh Kulkarni, Partha Pratim Pande, and Ananth Kalyanaraman, "Network-on-Chip Hardware Accelerators for Biological Sequence Alignment," *IEEE transactions on computers*/IEEE transactions on computers, vol. 59, no. 1, pp. 29–41, Jan. 2010, doi: <https://doi.org/10.1109/tc.2009.133>.
- [13] N. J. Curtis, K. E. Niemeyer, and C.-J. Sung, "Using SIMD and SIMT vectorization to evaluate sparse chemical kinetic Jacobian matrices and thermochemical source terms," *Combustion and flame*, vol. 198, pp. 186–204, Dec. 2018, doi: <https://doi.org/10.1016/j.combustflame.2018.09.008>.
- [14] M.-S. Yu, C.-Y. Yuan, T.-L. Chen, and J.-K. Lee, "Case Study: Optimization Methods with TVM Hybrid-OP on RISC-V Packed SIMD," *ResearchGate*, Jan. 2024. <https://doi.org/10.1109/ACCESS.2024.3397195>.
- [15] NVIDIA, "Whitepaper NVIDIA's Next Generation CUDA™ Compute Architecture: Fermi™ V1.1.", 2009, Available: https://www.nvidia.com/content/PDF/fermi_white_papers/NVIDIA_Fermi_Compute_Architecture_Whitepaper.pdf.
- [16] K. Karimi, N. Dickson, and F. Hamze, "A Performance Comparison of CUDA and OpenCL." Available: <https://arxiv.org/vc/arxiv/papers/1005/1005.2581v1.pdf>.
- [17] Y. Zhuo, T. Zhang, F. Du, and R. Liu, "A parallel particle swarm optimization algorithm based on GPU/CUDA," *Applied soft computing*, vol. 144, pp. 110499–110499, Sep. 2023, doi: <https://doi.org/10.1016/j.asoc.2023.110499>.
- [18] A. Khajeh-Saeed, S. Poole, and J. B. Perot, "Acceleration of the Smith-Waterman algorithm using single and multiple graphics processors," *Journal of Computational Physics*, vol. 229, no. 11, pp. 4247–4258, Jun. 2010, doi: 10.1016/j.jcp.2010.02.009.
- [19] Ranasinghe D. N. and Siriwardena T. R. P. "An efficient implementation of Smith Waterman algorithm on GPU using CUDA, for massively parallel scanning of sequence databases," *IEEE Conference Publication | IEEE Xplore*, May 01, 2009. <https://ieeexplore.ieee.org/abstract/document/5160931>.
- [20] X. Guo, H. Wang, and V. Devabhaktuni, "A systolic Array-Based FPGA parallel architecture for the BLAST algorithm," *ISRN Bioinformatics*, vol. 2012, pp. 1–11, Sep. 2012, doi: 10.5402/2012/195658.
- [21] P. D. Vouzis and N. V. Sahinidis, "GPU-BLAST: using graphics processors to accelerate protein sequence alignment," *Bioinformatics*, vol. 27, no. 2, pp. 182–188, Nov. 2010, doi: 10.1093/bioinformatics/btq644.
- [22] P. L. Miller, P. M. Nadkarni, and N. M. Carriero, "Parallel computation and FASTA: confronting the problem of parallel database search for a fast sequence comparison algorithm," *Bioinformatics*, vol. 7, no. 1, pp. 71–78, Jan. 1991, doi: 10.1093/bioinformatics/7.1.71.
- [23] Y. S. Lee, Y. S. Kim, and R. L. Uy, "Serial and parallel implementation of Needleman-Wunsch algorithm," *IJAIN (International Journal of Advances in Intelligent Informatics)*, vol. 6, no. 1, p. 97, doi: 10.26555/ijain.v6i1.361.
- [24] F. Muhamad, R. Ahmad, S. Asi, and M. Murad, "Performance analysis of Needleman-Wunsch Algorithm (Global) and Smith-Waterman Algorithm (Local) in reducing search space and time for DNA sequence alignment," *Journal of Physics. Conference Series*, vol. 1019, p. 012085, Jun. 2018, doi: 10.1088/1742-6596/1019/1/012085.

- [25] Lim, S. E., Lim, S. V., Ting, C. L., Wong, A. E., & Uy, R. L. (2021). SIMD Implementation of Local Alignment-Based Six-Frame Alignment Algorithm. *Philippine Computing Journal*, 17(1).